Classical Systems Biology

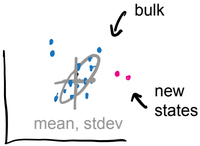
MCSB Bootcamp — Mathematical and Computational Track

Jun Allard

The goals, challenges, systems, tools and approaches of today

Data and analytics

Keystone **goal**: discover the hidden — new states, cell types, disease classes, ...



Keystone **challenge**: biological data is not just big, it is **high-dimensional**.

- 30000 genes, 1000 cells
- 10^4 microbial species, 20 patients

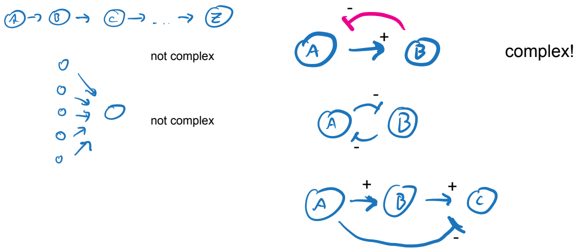
Keystone **tool**: clustering, regression, dimension-reduction ... coding.

Models and mechanism

Keystone **goal**: **prediction** — "what would happen if..."

if A is increased, then...

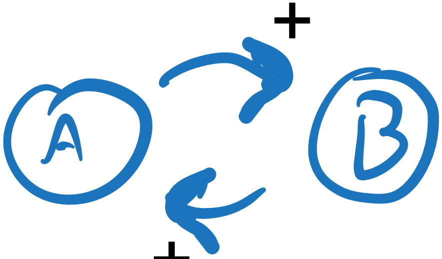
Keystone **challenge**: biological systems are **complex**, not just big.



Keystone **tool**: dynamical systems, control theory, (the differential equation,) ...

Most things in biology have feedback

stress



breathing rate

Complex systems and “systems thinking”

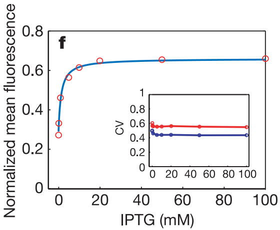
Early synthetic biology

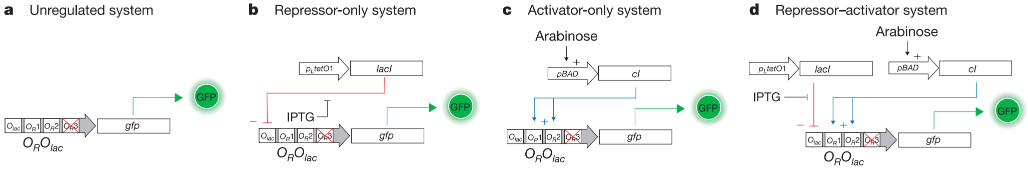
nature Vol 439|16 February 2006|doi:10.1038/nature04473

LETTERS

A bottom-up approach to gene regulation

Nicholas J. Guido[1]*, Xiao Wang[2]*, David Adalsteinsson[3], David McMillen[5], Jeff Hasty[6], Charles R. Cantor[1], Timothy C. Elston[4] & J. J. Collins[1]





"Computation drove the field of synthetic biology in its very early days."

Collins, 2020

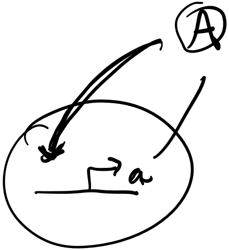
"We didn't have those data sets. We didn't have a wet lab. We didn't have the capability. We had to rely on mathematical modeling."

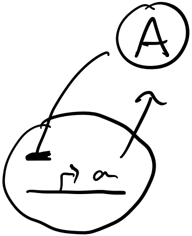
paraphrasing Collins, 2015

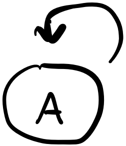
Complex biological systems and "systems thinking"

$\rightarrow$ Activation

$\dashv$ Repression







Positive feedback

With positive feedback, the curve showing steady-state protein as a function of activator TF should move up.

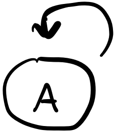
Does the curve move up by:

- shifting the EC50 to the left?
- moving up at high-C?
- moving up at low-C?
- a combination of all three?

Seven circuits

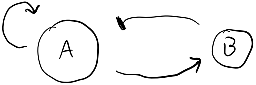
Activation

Repression



Positive feedback

1



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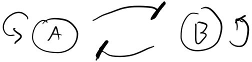

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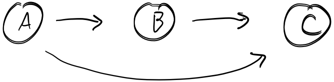

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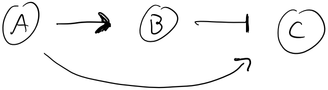

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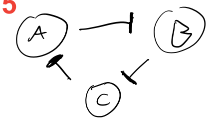

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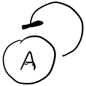

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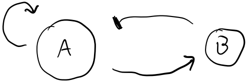
In-class activity: matching

1



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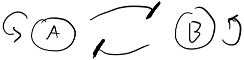

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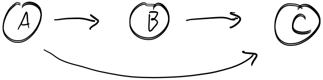

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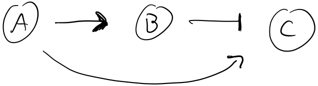

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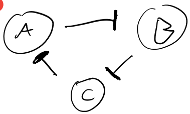

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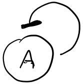

7

Match each circuit to its name.

- Direct negative feedback
- Indirect negative feedback
- Double-negative feedback
- Fast positive and slow negative feedback
- Incoherent feedforward
- Coherent feedforward
- The Repressilator